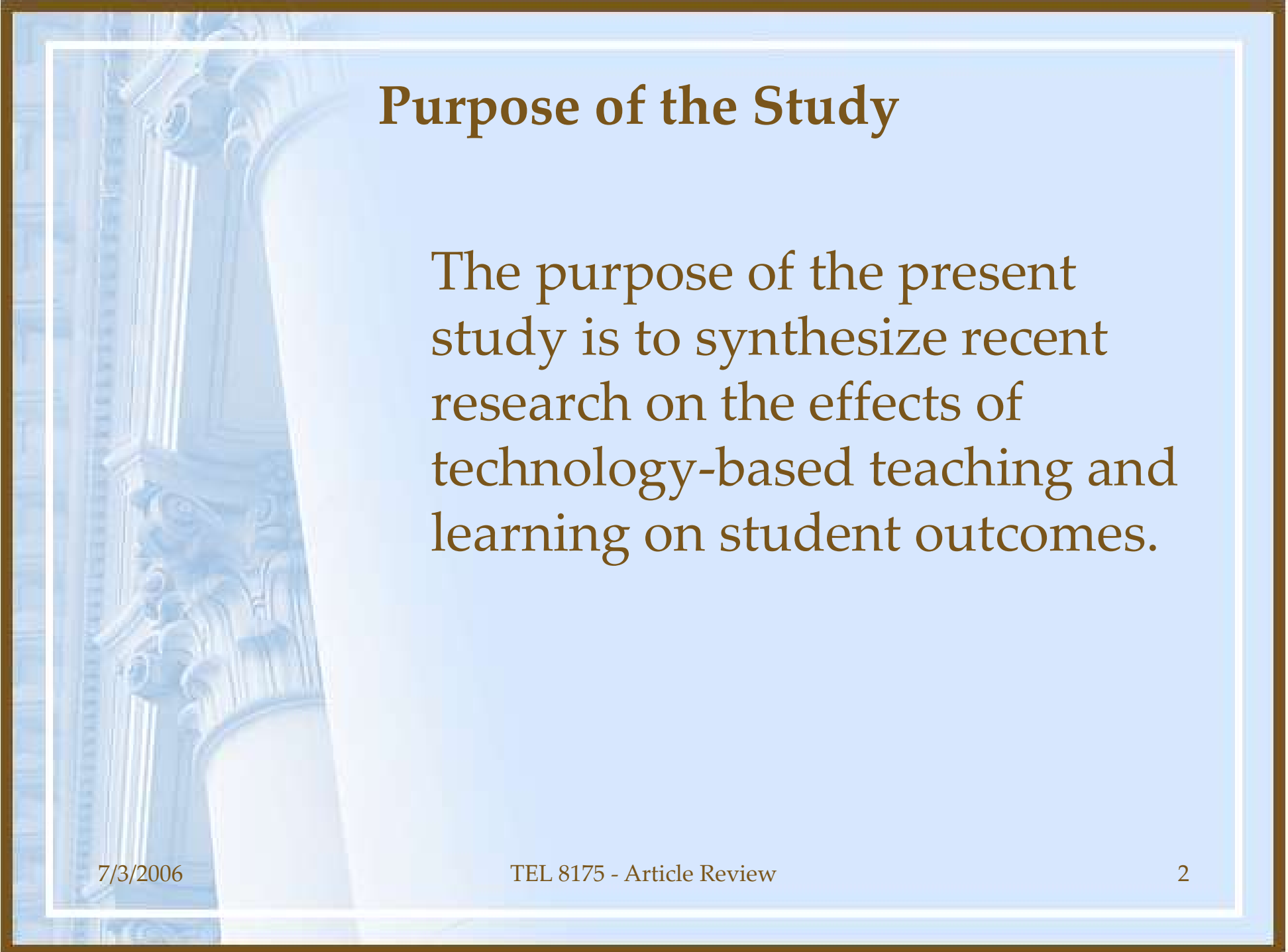


The background of the slide features a light blue gradient. On the left side, there is a vertical image of classical columns, possibly from a government building, rendered in a lighter blue tone. The entire slide is framed by a thin brown border.

# Quantitative Synthesis of Technology

Jim Pham

The background of the slide features a faint, light blue image of classical architectural columns, possibly from a library or university building, which are partially visible on the left side.

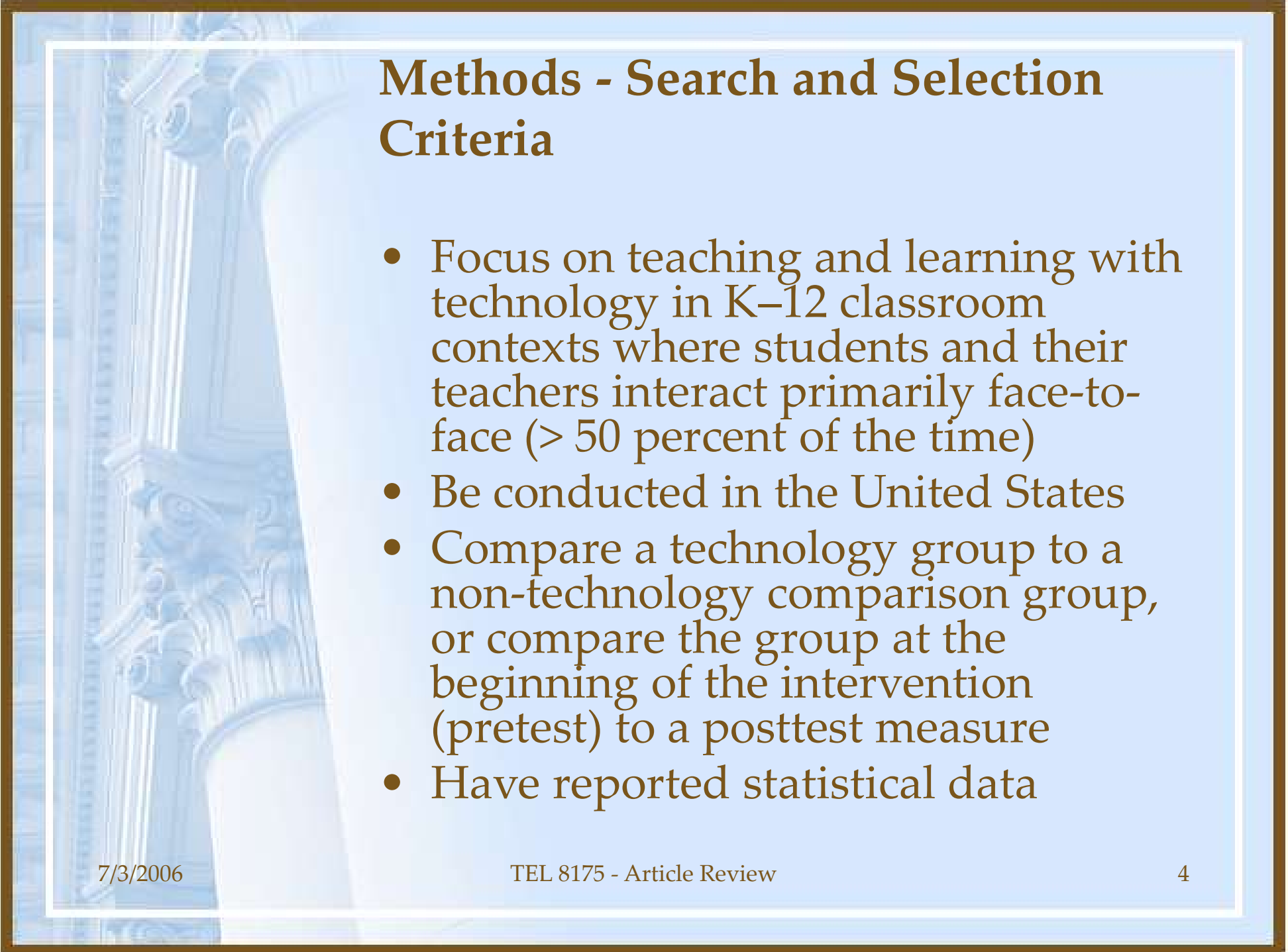
## Purpose of the Study

The purpose of the present study is to synthesize recent research on the effects of technology-based teaching and learning on student outcomes.

The background of the slide features a light blue gradient with a faint, vertical image of classical columns on the left side. The columns are white with detailed capitals and fluted shafts.

## Focus on three areas of the outcome learning

- Cognitive
- Affective
- Behavioral



## Methods - Search and Selection Criteria

- Focus on teaching and learning with technology in K–12 classroom contexts where students and their teachers interact primarily face-to-face (> 50 percent of the time)
- Be conducted in the United States
- Compare a technology group to a non-technology comparison group, or compare the group at the beginning of the intervention (pretest) to a posttest measure
- Have reported statistical data

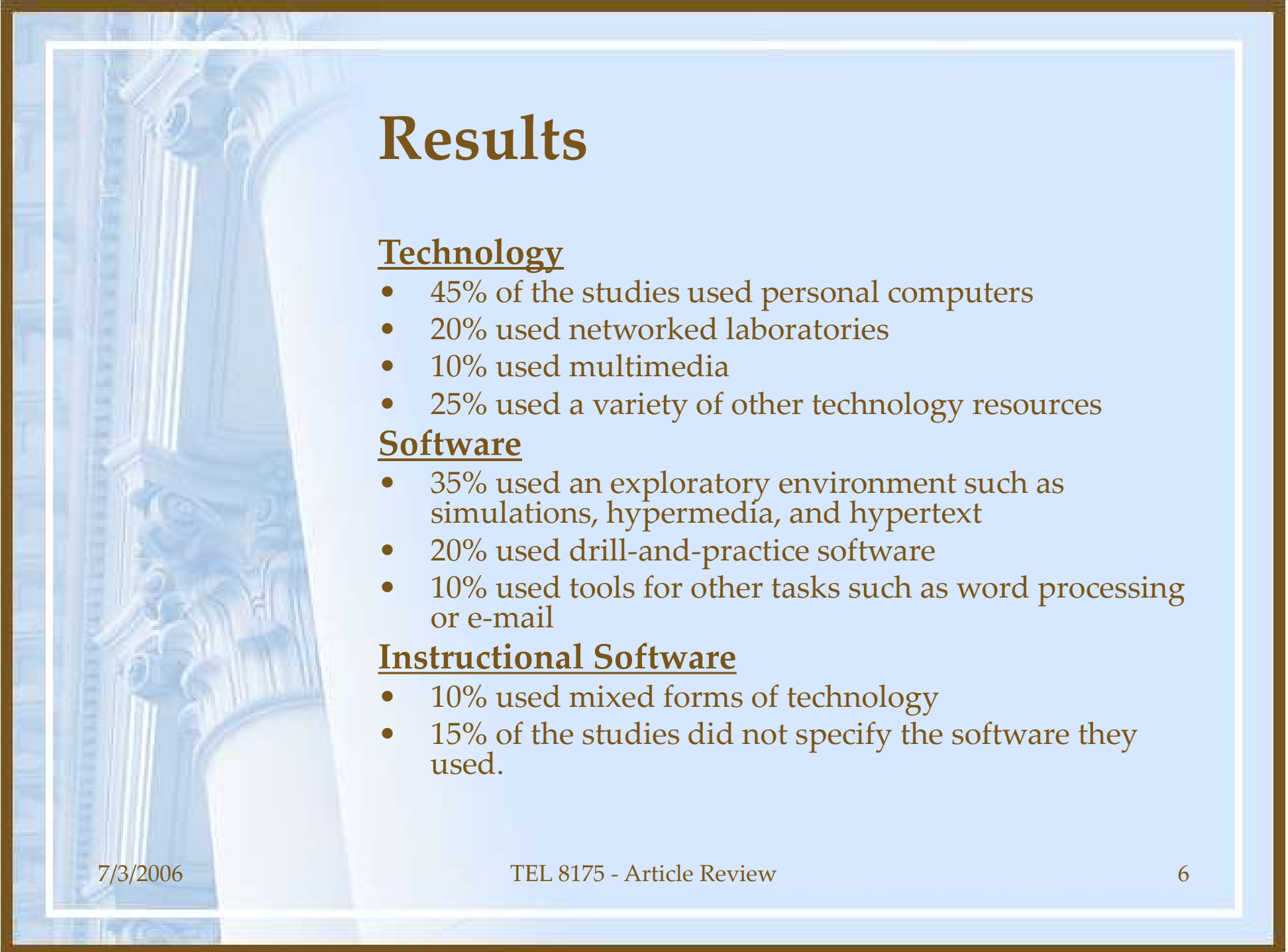
# Procedure

- To calibrate the studies' results, or place them on a common scale, effect sizes were calculated.
- These effect sizes consist of the treatment group means minus the control mean divided by the control standard deviation.
- Effect sizes can be considered a standardized estimate of where the treatment group stands in comparison with the control group distribution.

## Formula

$$1/n$$

where  $n$  = number of comparisons in the study



# Results

## Technology

- 45% of the studies used personal computers
- 20% used networked laboratories
- 10% used multimedia
- 25% used a variety of other technology resources

## Software

- 35% used an exploratory environment such as simulations, hypermedia, and hypertext
- 20% used drill-and-practice software
- 10% used tools for other tasks such as word processing or e-mail

## Instructional Software

- 10% used mixed forms of technology
- 15% of the studies did not specify the software they used.

## Results (cont.)

**Table 2a**  
**Summary of Mean Study-Weighted Effect Sizes for Student Outcomes**

| <b>Outcomes</b> | <b>Number of<br/>Weighted<br/>Comparisons</b> | <b>Study-<br/>Weighted<br/>Effect Sizes</b> | <b>SD</b>   | <b>95% Confidence<br/>Intervals</b> |               |
|-----------------|-----------------------------------------------|---------------------------------------------|-------------|-------------------------------------|---------------|
|                 |                                               |                                             |             | <b>Lower</b>                        | <b>Higher</b> |
| Cognitive       | 13                                            | .390                                        | .711        | -.050                               | .830          |
| Affective       | 5                                             | .279                                        | .493        | -.305                               | .862          |
| Behavioral      | 2                                             | -.154                                       | .402        | -2.503                              | 2.195         |
| <b>Overall</b>  | <b>20</b>                                     | <b>.301</b>                                 | <b>.633</b> | <b>.004</b>                         | <b>.598</b>   |

## Results (cont.)

**Table 2b**  
**Summary of Unweighted Effect Sizes for Student Outcomes**

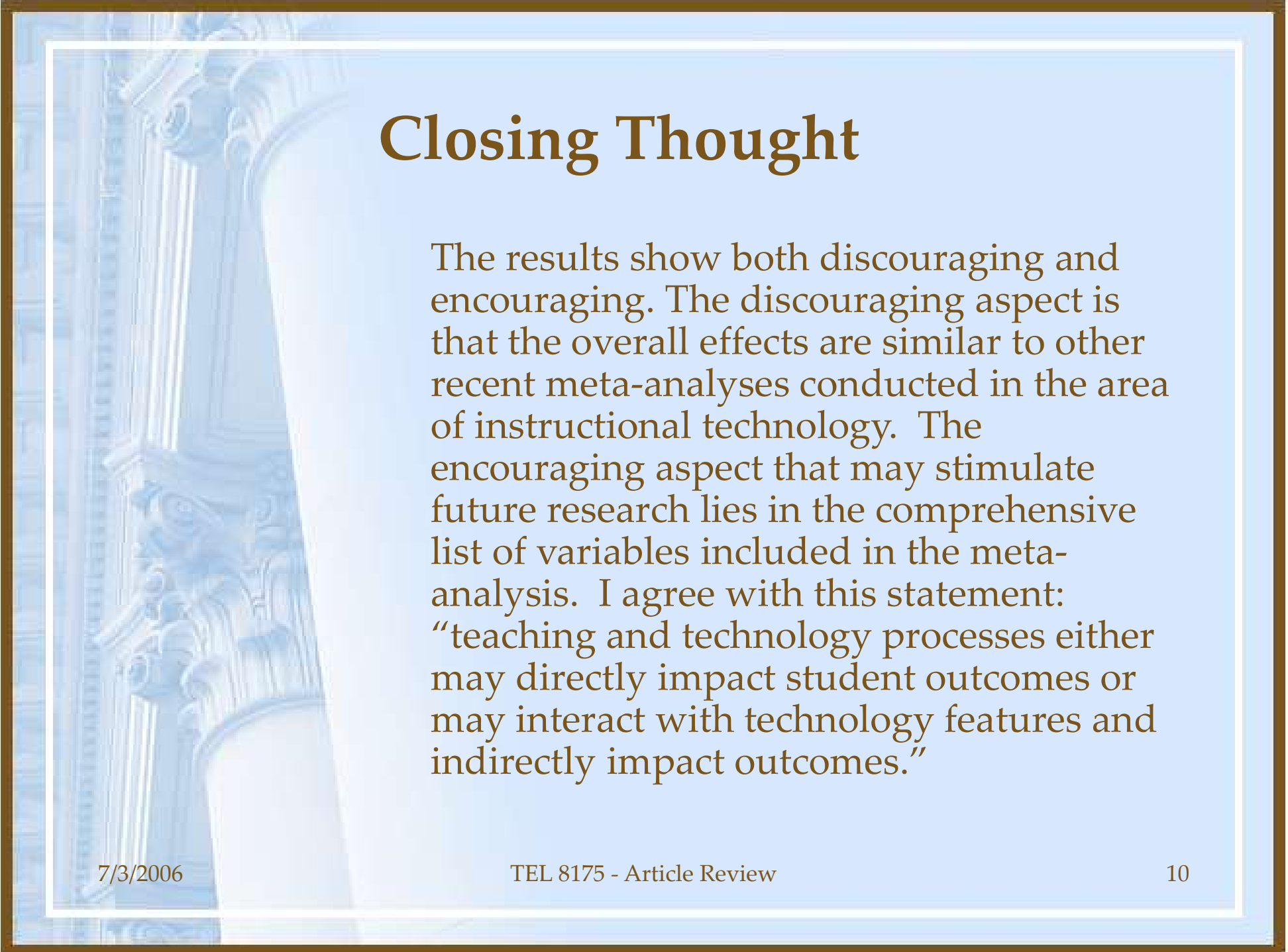
| <b>Outcomes</b> | <b>Number of<br/>Unweighted<br/>Comparisons</b> | <b>Unweighted<br/>Effect Sizes</b> | <b>SD</b>   | <b>95% Confidence<br/>Intervals</b> |               |
|-----------------|-------------------------------------------------|------------------------------------|-------------|-------------------------------------|---------------|
|                 |                                                 |                                    |             | <b>Lower</b>                        | <b>Higher</b> |
| Cognitive       | 48                                              | .327                               | .589        | .156                                | .498          |
| Affective       | 60                                              | .208                               | .397        | .106                                | .311          |
| Behavioral      | 30                                              | .096                               | .235        | -.002                               | .174          |
| <b>Overall</b>  | <b>138</b>                                      | <b>.223</b>                        | <b>.455</b> | <b>.146</b>                         | <b>.229</b>   |



# Other recent Meta-analyses in technology

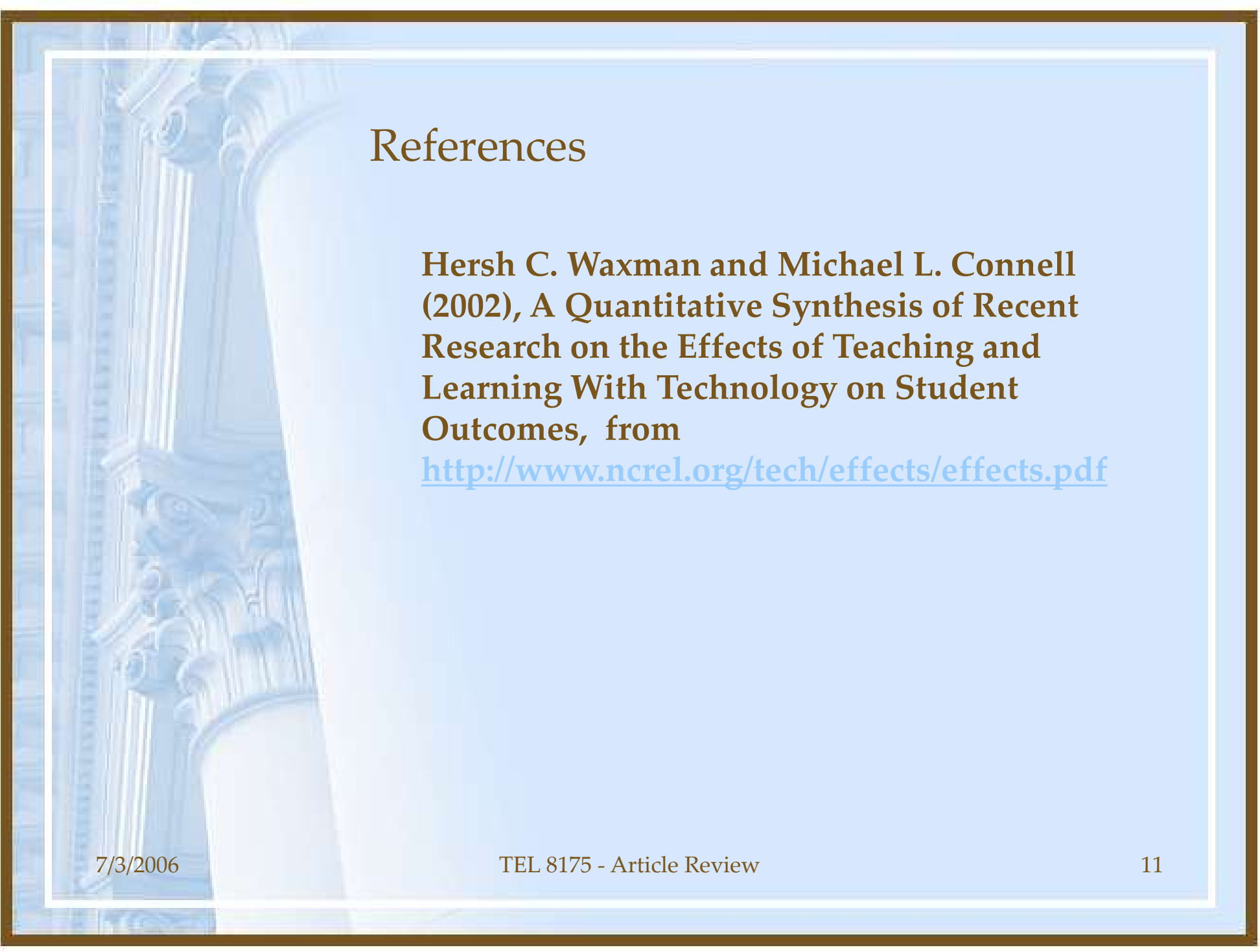
**Table 1**  
**Summary of Recent Meta-Analyses in Educational Technology**

| Author(s) and Date                        | Focus                                                         | N of Studies | Effect Size |
|-------------------------------------------|---------------------------------------------------------------|--------------|-------------|
| Bayraktar (2001-2002)                     | CAI in secondary and college science                          | 42           | .273        |
| Blok, Oostdam, Otter, and Overmaat (2002) | Computer-based instructional simulations                      | 42           | .190        |
| Cavanaugh (2001)                          | Interactive distance education technologies                   | 19           | .147        |
| Christmann and Badgett (1999)             | CAI in science                                                | 11           | .266        |
| Christmann, Badgett, and Lucking (1997)   | CAI in differing subject areas                                | 27           | .209        |
| Christmann, Lucking, and Badgett (1997)   | CAI in secondary schools                                      | 28           | .172        |
| Lee (1999)                                | Computer-based instructional simulation                       | 19           | .410        |
| Lou, Abrami, and d'Apollonia (2001)       | Small group v. individualized learning with technology        | 122          | .150        |
| Whitley (1997)                            | Gender differences in computer-related attitudes and behavior | 82           | .209        |
| Median =                                  |                                                               | 28           | .209        |

The background of the slide features a faint, blue-tinted image of classical architectural columns, likely from a Greek or Roman temple, positioned on the left side. The columns are fluted and have ornate capitals. The overall background is a light blue gradient.

# Closing Thought

The results show both discouraging and encouraging. The discouraging aspect is that the overall effects are similar to other recent meta-analyses conducted in the area of instructional technology. The encouraging aspect that may stimulate future research lies in the comprehensive list of variables included in the meta-analysis. I agree with this statement: “teaching and technology processes either may directly impact student outcomes or may interact with technology features and indirectly impact outcomes.”

The background of the slide features a light blue gradient with a faint, vertical image of classical columns on the left side. The columns are white with detailed capitals and are set against a darker blue background.

## References

**Hersh C. Waxman and Michael L. Connell  
(2002), A Quantitative Synthesis of Recent  
Research on the Effects of Teaching and  
Learning With Technology on Student  
Outcomes, from  
<http://www.ncrel.org/tech/effects/effects.pdf>**